

Practical #6

Practical Title:

To deploy and manage a simple web application using version control and hosting techniques

Theory:

1. What is Web Hosting?

Web hosting is a service that allows you to store your website files on a server and make them accessible on the internet.

Example: When you open a website, files are loaded from a hosting server.

2. What is GitHub Hosting (GitHub Pages)?

GitHub Pages is a free hosting service provided by GitHub.

- Used to host **static websites (HTML, CSS, JS)**
 - Provides a live URL for your project
 - Automatically updates when you push code
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3. Role of Version Control in Deployment

Version control (Git) helps in:

- Tracking changes in code
- Managing different versions
- Easy rollback if errors occur
- Smooth deployment using GitHub integration

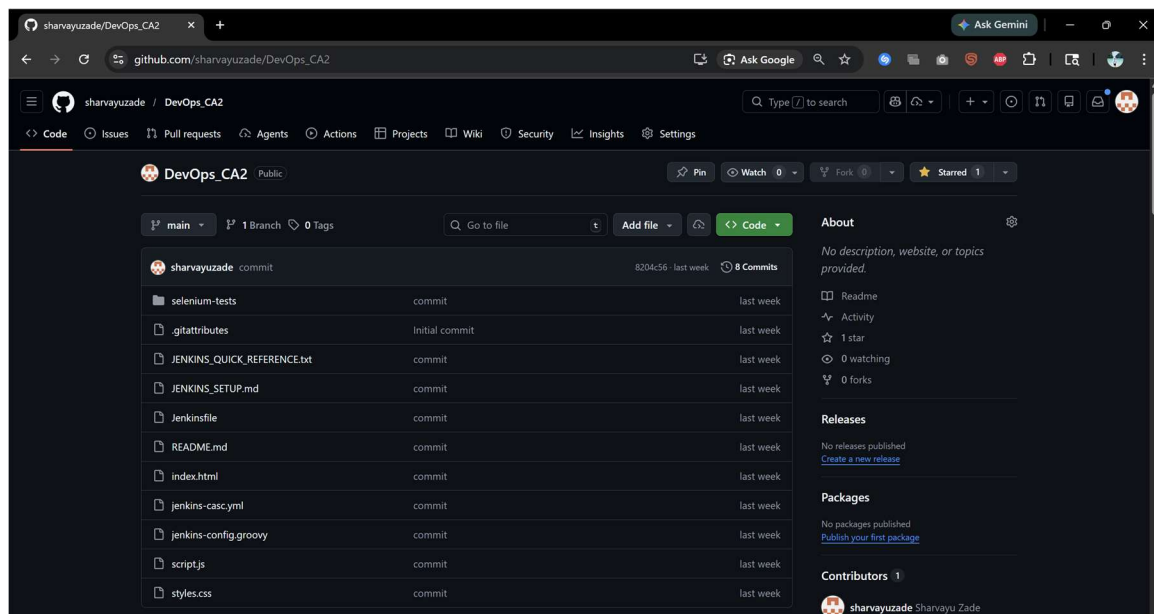
Procedure:

Step 1: Create a simple web application (index.html)

Step 2: Initialize Git repository

- Step 3: Add and Commit files
- Step 4: Create GitHub repository
- Step 5: Push code to GitHub
- Step 6: Enable GitHub Pages
- Step 7: Access deployed website using provided link

Output Screenshots:



Activity / Task:

Task 1: Deploy Web Application

Repository Link: https://github.com/sharvayuzade/DevOps_CA2

Live Website Link: https://sharvayuzade.github.io/DevOps_CA2

Task 2: Update Website

Commands Used:

```
git add .  
git commit -m "Updated website"  
git push origin main
```

After pushing, GitHub Pages updates automatically.

Task 3: Benefits of Using GitHub for Deployment

- Free hosting
 - Easy integration with Git
 - Automatic updates
 - Version control + deployment in one platform
 - Good for beginners and small projects
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Advantages

1. Free hosting service
 2. Easy deployment process
 3. Automatic updates
 4. Integrated with Git
 5. Supports collaboration
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Challenges

1. Only static websites supported
2. Limited customization
3. Requires Git knowledge